Day-18 | Quantum Finance

Portfolio Optimization & Risk Analysis

1. Learning Objectives

- ✓ Understand what Quantum Finance is and its core applications.
- ✓ Explain the computational challenge of combinatorial explosion in finance.
- ✓ Model portfolio optimization mathematically using risk and return.
- ✓ Build intuition for using parameterized quantum circuits to explore financial portfolios.

2. Quick Recap

- **Quantum Optimization:** Finding the best solution from vast possibilities using hybrid loops.
- **Parameterized Circuits:** Adjusting gate angles to shift measurement probabilities toward better solutions.

Transition: Today, we apply these exact optimization techniques to real-world financial decision-making, where finding the optimal balance of risk and return is critical.

3. Why Do We Need Quantum Finance?

Motivation: Financial institutions must choose the best combination of assets from thousands of options. With just 100 stocks (Buy/Don't Buy), there are 2^{100} possible portfolios. This combinatorial explosion makes exhaustive classical searching impossible. Quantum optimization aims to navigate this massive search space intelligently.

Core Applications in Finance

- **Portfolio Optimization:** Balancing maximum return with minimum risk.
- **Risk Analysis:** Estimating potential losses across millions of simulated market scenarios.
- **Derivative Pricing:** Faster calculation of complex options contracts.
- **Fraud Detection:** Using QML to find anomalous transaction patterns in massive datasets.

4. Core Mathematical Representation: The Portfolio

Binary Decision Variables

$$x_i \in \{0, 1\}$$

Where:

- $x_i = 1 \implies$ Invest in asset i.
- $x_i = 0 \implies$ Do not invest in asset i.
- *Intuition:* A 4-asset portfolio like $[1, 0, 1, 1]$ maps perfectly to a 4-qubit computational basis state 1011.

5. Concept Expansion: Myths vs. Reality

Common Misconception	Reality / Correction
"Quantum computers predict stock prices."	**Correction:** They don't predict the future; they solve complex mathematical optimization and simulation problems faster.
"Quantum finance replaces classical finance."	**Correction:** The financial theory (risk vs. return) remains identical. The *computational engine* evaluating the theory changes.

6. Core Mechanism: Balancing Risk and Return

The objective of portfolio optimization is to minimize the "Cost Function".

The Cost Function ($C(x)$):

$$C(x) = \text{Risk}(x) - \lambda \cdot \text{Return}(x)$$

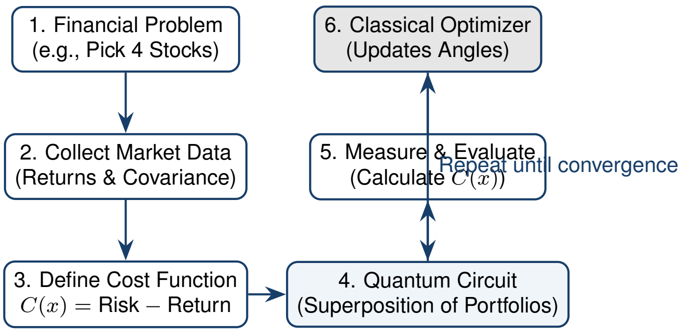
Where:

- $\text{Risk}(x) = x^T Q x$ (calculated using the covariance matrix Q).
- $\text{Return}(x) = \sum_{i=1}^{N} r_i x_i$ (sum of expected returns for chosen assets).
- λ is a weight parameter balancing the investor's risk tolerance.

7. Visualizing Portfolio Choices

Portfolio x	Return	Risk	Optimizer Evaluation
1010	12%	Low	Good, safe choice.
1100	16%	High	High return, but too risky.
0111	10%	Medium	Sub-optimal balance.
1011	**24%**	**Medium**	**Optimal Target (Best Balance)**

8. Workflow: From Finance to Quantum Circuit



9. Practical Implementation: Preparing Candidate Portfolios

This Qiskit snippet sets up a parameterized circuit where 4 qubits represent 4 assets. Adjusting the R_y angles changes the probability of choosing each asset.

```python
from qiskit import QuantumCircuit
from qiskit.primitives import StatevectorSampler

# 1. Initialize 4 Qubits (representing 4 stocks)
qc = QuantumCircuit(4, 4)

# 2. Create superposition of all possible portfolios
qc.h(range(4))

# 3. Parameterized Rotations (The "weights" updated by an optimizer)
# These angles bias the probability of measuring 0 or 1 for each stock
qc.ry(0.8, 0)
qc.ry(1.2, 1)
qc.ry(0.5, 2)
qc.ry(1.0, 3)

# 4. Measure the candidate portfolio
qc.measure(range(4), range(4))

# 5. Execute
sampler = StatevectorSampler()
job = sampler.run([qc], shots=1024)
counts = job.result()[0].data.meas.get_counts()
print(counts) # e.g., {'1011': 248, '1110': 215, ...}
```

10. ⋆ Key Takeaways

- > **Quantum Finance** applies quantum optimization to handle the combinatorial explosion of financial choices (like 2^{100} possible portfolios).
- > Investment decisions map perfectly to qubits: 1 (Invest) and 0 (Do not invest).
- > The goal is to minimize a **Cost Function** that mathematically balances Expected Return against Portfolio Risk.
- > **Parameterized Quantum Circuits** explore these possibilities, with classical optimizers iteratively guiding the system toward the optimal portfolio state.